

# Hybrid Retrieval and Neural Aggregation for Climate Change Verification

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## Abstract

Automated fact verification of climate-related claims is challenging because evidence combines scientific facts with interpretation and frequently contradicts across sources. We propose a hybrid retrieval and neural aggregation framework for the CLIMATE-FEVER task. The pipeline retrieves the top-100 candidates with BM25 over a curated 3,443-passage pool, re-ranks them with a transformer cross-encoder, and assigns a claim-level label from the top-5 retrieved sentences. We compare two aggregation strategies: an NLI cross-encoder with rule-based consistency analysis that explicitly targets the DISPUTED class identified by Diggelmann et al. (2021) as future work, and a BiLSTM classifier over frozen [CLS] embeddings from pre-trained encoders (BERT, RoBERTa, DeBERTa-v3). Our best configuration, RoBERTa-base with a BiLSTM aggregator, achieves 53.90% dev accuracy, an 18.84-point absolute improvement over the specification-sheet baseline of 35.06%. The retrieval stage displays an F-score of 0.2465. Per-class analysis shows the model handles SUPPORTS and REFUTES reasonably ( $F1 = 0.66$  and  $0.50$  respectively) but never predicts DISPUTED ( $F1 = 0.00$ ), which states that handling of conflicting evidence remains an open problem.

## 1 Introduction

In recent years, automated fact verification systems have become an active area of exploration. The spread of climate-related misinformation raises serious concerns for policy formation, scientific understanding, and public literacy. These climate-related claims are a unique case because of how the evidence is presented. It combines scientific facts with interpretation and differing perspectives among sources, which often contradict one another.

Our aim is to propose a model capable of explicit evidence-level reasoning. The pipeline uses BM25

for fast retrieval of candidate evidence, then, using a transformer cross-encoder, it re-ranks them to the top 5. The model can categorize them into three labels: supports/refutes/evidence agree, disputed (evidence conflicts), and not enough info.

### 1.1 Literature Review

The data set was derived from a paper proposed by Thorne et al. (2018), who introduced the available data set for text verification against the evidence sources, FEVER. It was created by human annotators, who manually verified 185,445 claims against the first section of Wikipedia and separately annotated them as trustworthy evidence from multiple pages and sentences, based on the appropriate baseline and guides. They constructed a simple pipeline that comprised: document retrieval, sentence-level selection using TF-IDF similarity for claim, and they compared the MLP baseline and the Decomposable Attention (DA) model. Their complete pipeline shows a moderately final test result of 31% correctly classified evidence and 50.91% accuracy abandoning the evidence.

Prior work by Diggelmann et al. (2021) developed a pipeline approach for evidence retrieval and a claim validation system similar to the baseline system proposed by FEVER. Their algorithm automatically retrieves ALBERT sentence embeddings that retrieve semantically similar evidence candidates. Then it ranks the evidence using a neural model and predicts whether the evidence supports, refutes, or is insufficient to support the claim. However, unlike the earlier paper, CLIMATE-FEVER focused on data collected from Google Searches, containing 1,535 climate-related claims, paired with 7,675 annotated claim-evidence pairs retrieved from Wikipedia. Their contribution is the introduction of DISPUTED claims that support and refute evidence. Their main findings show that the FEVER model performed poorly on CLIMATE-FEVER, achieving 39% accuracy.

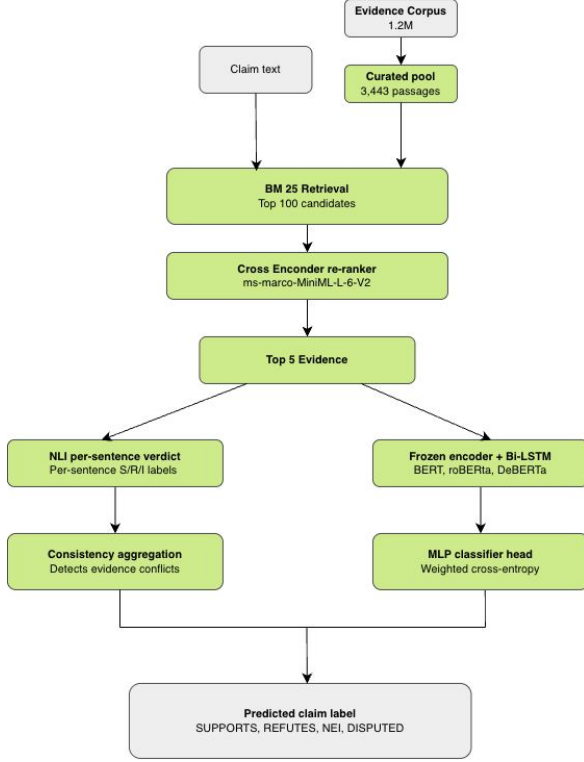


Figure 1: System Architecture Framework.

The above works provide the context of the needs model that enables evaluation of LLM on verification checks for the climate-related verification task. The key gap in existing prior systems was that they limited evidence-level reasoning and could not explicitly analyze the reasons for sentence classification. Other than that, weak handling of evidence that contradicts and struggles with the ambiguity of data, such as partially agreeing on whether the evidence is contradictory or supportive, which leads to a lack of explainability, as the existing pipeline is unable to capture the complexity of numerical inconsistencies and ambiguity.

This study explores automated fact verification using NLP to retrieve textual evidence for a given claim and classify it into three categories: supported, refuted, or insufficient evidence. We addressed the gap by proposing a fine-grained evidence-reasoning framework that retrieves candidate evidence using BM25 and transformers before the aggregation stage, and analyzes evidence consistency and conflict patterns to directly influence the final labels of the claims. This enables a more interpretable and robust system to handle conflicting real-world climate evidence.

| Split        | #Claims | #Evidence (corpus) |
|--------------|---------|--------------------|
| Train        | 1,228   | —                  |
| Dev          | 154     | —                  |
| Test         | 153     | —                  |
| Full corpus  | —       | 1,208,827          |
| Curated pool | —       | 3,443              |

Table 1: Dataset statistics for CLIMATE-FEVER.

| Label           | Train | %    | Dev | %    |
|-----------------|-------|------|-----|------|
| SUPPORTS        | 519   | 42.3 | 68  | 44.2 |
| REFUTES         | 199   | 16.2 | 27  | 17.5 |
| NOT_ENOUGH_INFO | 386   | 31.4 | 41  | 26.6 |
| DISPUTED        | 124   | 10.1 | 18  | 11.7 |

Table 2: Class distribution across splits.

## 2 Approach

The proposed framework combines a hybrid two-stage retrieval pipeline with transformer-based semantic re-ranking to identify relevant passages from a full-text corpus of our data as shown in Table 1. The baseline system achieves a retrieval F-score of 0.338, classification accuracy of 0.351, and harmonic mean of 0.344 on the development set.

Our system starts by using the BM25 (Robertson and Zaragoza, 2009) algorithm to retrieve the top 100 candidate evidence passages for each claim. In Figure 1.

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) (k_1 + 1)}{f(t, d) + k_1 K(d)}, \quad (1)$$

where  $K(d) = 1 - b + b |d| / \text{avgdl}$ ,  $f(t, d)$  is the frequency of term  $t$  in document  $d$ ,  $|d|$  is the document length,  $\text{avgdl}$  is the average document length across the corpus, and  $k_1$  and  $b$  are hyperparameters controlling term-frequency saturation and document length normalization.

BM25 calculates the correlation between passages and their ability to rank documents with overlapping keywords, and accounts for term-frequency differences between the claim and the evidence. Moreover, to improve efficiency, a curated evidence pool was constructed comprising 3,443 evidence passages derived from the training and development datasets. While BM25 is effective for lexically similar passages, transformers can capture deeper semantic relationships between claims and evidence. The re-ranker was applied after retrieval

to polish evidence passages based on semantic relevance.

Another novel approach is to introduce a sentence-level evidence reasoning stage prior to final claim classification. The architecture uses an NLI cross-encoder (nli-MiniLM2-L6-H768) (Wang et al., 2021) to capture the semantic relationship between the text claim and evidence. Hence, instead of directly predicting the claim label from aggregated evidence, the sentence is classified into one out of three classes for mapping: SUPPORT, REFUTE, AND IRRELEVANT. The SUPPORTS label indicates that the evidence semantically agrees with the claim, whereas REFUTES shows that the evidence contradicts the claim. An additional label, NEUTRAL, is added when the claim does not have enough evidence to support or refute the claim.

We set a threshold of 0.1 to determine the classification confidence. This aggregation mechanism is particularly important for handling complex claims containing contradictory or mixed evidence. The DISPUTED class, in particular, is identified by Diggelmann et al. (2021) as future work because no standard NLI formulation captures both-sides reasoning. Therefore, we implemented the framework in a stage that enables analysis of the evidence sentence individually, thereby improving evidence reasoning before the prediction stage.

Alongside the rule-based consistency aggregator, we implemented another, learned aggregation part which predicts the labels based on the earlier retrieved top-5 evidence sentences. This essentially is a four-way supervised classification problem.

We encode each claim and each retrieved evidence by a pre-trained transformer whose weights we keep frozen. We chose the frozen models for two main reasons. First, it reduces computation: every embedding can be pre-computed once and reused across all training epochs. Second, it prevents a large encoder from overfitting which in the results we will see still remains an issue. After doing research on the models used in similar scenarios we chose three encoders, all with hidden size 768: BERT-base-uncased (Devlin et al., 2019), RoBERTa-base (Liu et al., 2019), and DeBERTa-v3-base (He et al., 2023). For BERT and RoBERTa, we take the [CLS] token at position 0 as the pooled sentence representation. For DeBERTa-v3 we instead apply mean pooling over all non-padding tokens, because its disentangled-attention design means the [CLS] token does not accumulate a us-

able pooled summary without task-specific fine-tuning. All inputs are truncated to a maximum length of 128 tokens. Each claim is therefore represented by a single 768-dimensional vector, and its evidence set by a variable-length sequence of 768-dimensional vectors. One additional approach we tried was to unfreeze the top 2, 4 and 6 layers of the encoders to evaluate the performance. However, that resulted in a much slower training and also in worse accuracy interestingly, most likely the result of overfitting since we have a very small training dataset.

The variable-length sequence of evidence embeddings is summarised by a recurrent network. We compare a bidirectional LSTM (BiLSTM) against a unidirectional LSTM (UniLSTM) baseline in order to test whether reading the evidence sequence in both directions improves verification. The evidence embeddings are packed before being passed to the LSTM so that padding does not affect the recurrence. For the BiLSTM, the final forward and backward hidden states of the top layer are concatenated into a 512-dimensional evidence summary; for the UniLSTM, only the final forward hidden state (256-dimensional) is used. This evidence summary we concatenate with the 768-dimensional claim embedding and passed through a two-layer feed-forward classifier (Linear  $\rightarrow$  ReLU  $\rightarrow$  Dropout  $\rightarrow$  Linear) which produces the four class logits.

## 3 Experiments

### 3.1 Data

We evaluate on the CLIMATE-FEVER dataset introduced by Diggelmann et al. (2021), which contains 1,535 real-world climate-related claims paired with evidence retrieved from Wikipedia. We use the provided train/dev/test split of 1,228/154/153 claims respectively, against a full evidence corpus of 1,208,827 passages (Table 1). The dataset exhibits significant class imbalance: SUPPORTS dominates at 42.3% of training claims, while DISPUTED accounts for only 10.1% (Table 2). This imbalance directly motivates our use of inverse-frequency class weights in the loss function. To improve retrieval efficiency and quality, we constructed a curated evidence pool of 3,443 passages by filtering the full corpus to only those passages that appear as gold evidence in the training and development sets. This ensures the retrieval stage operates over climate-specific, high-relevance pas-

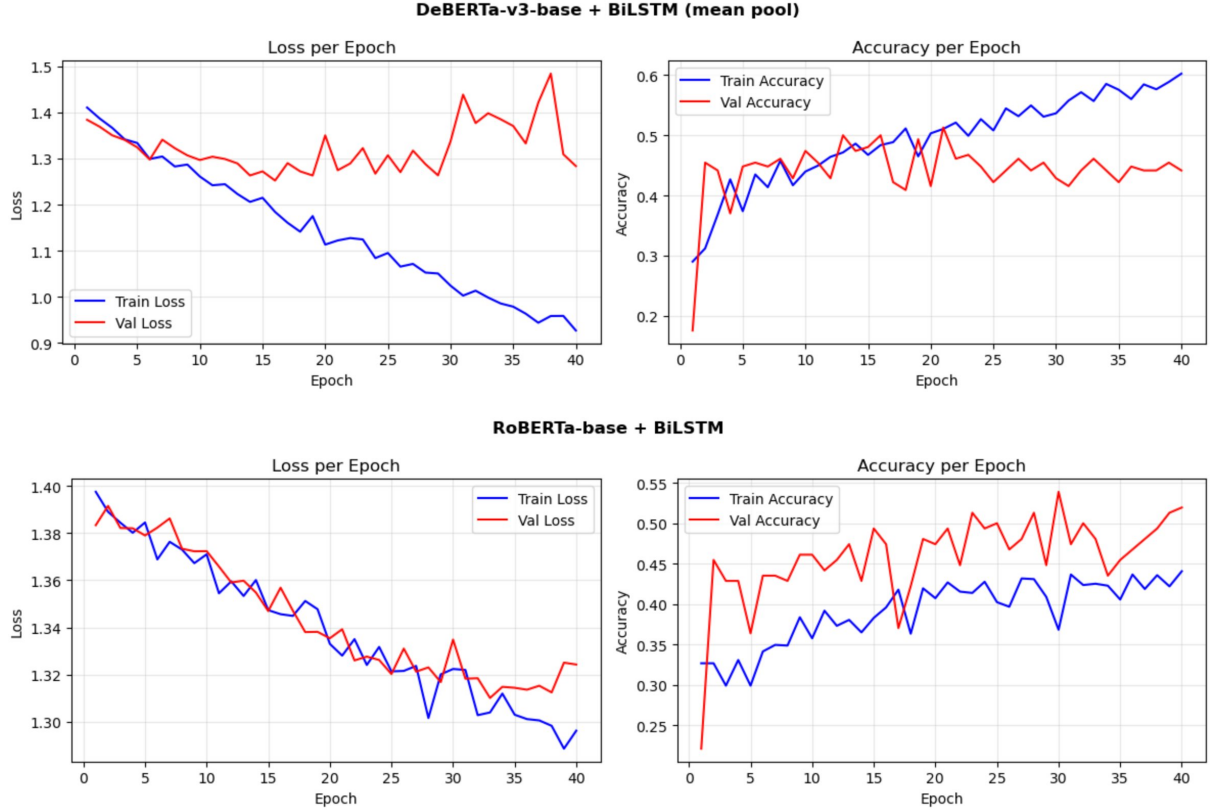


Figure 2: Training curves for DeBERTa-v3-base + BiLSTM (top) and RoBERTa-base + BiLSTM (bottom).

sages rather than the noisy full corpus.

### 3.2 Evaluation Metrics

In our evaluation, we evaluate both quantitative metrics and human diagnostics checks that cover the retrieval quality and evidence-based quality.

In our automatic qualitative, for each claim we compare the top 5 retrieved evidence set against the gold evidence. We compute the precision, recall, and F1,

$$P = \frac{|G \cap R|}{|R|}, \quad (2)$$

$$R = \frac{|G \cap R|}{|G|}, \quad (3)$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}, \quad (4)$$

where  $G$  is the gold evidence set and  $R$  is the top-5 retrieved set, and get the macro average F1 across all claims (reported in Table 5). To observe the classification accuracy, we gather claims whose predicted label exactly matches the gold label across for classes: (SUPPORTS, REFUTES, NOT\_ENOUGH\_INFO, DISPUTED). This is defined by a fraction of dev claims whose predicted claim\_label exactly matches the gold label across

the four-way label set. We also break this down per class in Table 4 so that minority-class behaviour is not hidden inside the overall accuracy figure.

We compute harmonic mean by combining our retrieval f-score and classification accuracy into a single score,

$$H = \frac{2 \cdot F \cdot A}{F + A}, \quad (5)$$

where  $F$  is the retrieval F-score and  $A$  is the classification accuracy.

For those who can't be scored we check per sentence checker, by printing each sentence with NLI verdict and softmax probabilities. By applying this, we can define whether the model is able to understand the logic even when the final output label is false.

In addition, we compare the distribution of the predicted labels to train and dev. It displays that heavy skew toward one class (e.g. NOT\_ENOUGH\_INFO) flags class-imbalance leakage even when accuracy looks fine.

DISPUTED scoring 0.00 on the per-class report is worth flagging as a known limitation, which we discuss further in Section 3.5.



### 3.3 Experimental Setup

For our experiment, we construct BM25 via rank\_bm25 (BM25Okapi) by applying lowercase and punctuation strip with a curated pool of 3,443 passages. We implemented the re-ranker with cross-encoder ms-marco-MiniLM-L-6-v2 with the max length of 512. For NLI, we used cross-encoder/nli-MiniLM2-L6-H768 (Wang et al., 2020) with a threshold of 0.10.

We compared three frozen pre-trained encoders — BERT-base-uncased, RoBERTa-base, and DeBERTa-v3-base — each with hidden size 768. In every case the encoder weights are kept fixed and run in evaluation mode, so no gradients flow into the encoder; embeddings are pre-computed once and cached, which removes the encoder from the training loop and ensures every epoch reuses identical features. All inputs are truncated to a maximum of 128 tokens. For BERT and RoBERTa we take the [CLS] token at position 0 of the final hidden layer as the pooled 768-dimensional representation. DeBERTa-v3 is the exception: its disentangled-attention design means the [CLS] token does not accumulate a usable pooled summary without task-specific fine-tuning, so we instead apply mean pooling over all non-padding tokens. This produces a 768-dimensional vector per claim and per evidence sentence regardless of encoder, keeping the downstream aggregator identical across all three. In order to find out the best parameters for UniLSTM and BiLSTM we performed an extensive GridSearch and evaluated the accuracy. That was done in order to balance accuracy and avoid overfitting as much as possible. In the end BiLSTM uses hidden\_dim=128, num\_layers=2, dropout=0.4, and num\_classes=4 (4,007,172 trainable params), while the UniLSTM has the same shape but is unidirectional (1,840,388 trainable params). The optimiser is Adam with lr=1e-3 and weight\_decay=1e-4, and the loss is CrossEntropyLoss with inverse-frequency class weights. We use a batch size of 32, 15 epochs, and grad clip 1.0.

### 3.4 Results

The claim-classification accuracy for every model configurations on development set is reported in Table 3. The results of our proposed framework shows that the current choice of recurrent aggregator turned out to have quite a significant effect. Using BERT-base to compare and make the choice, de-

| Model                              | Retrieved Dev | Gold Dev |
|------------------------------------|---------------|----------|
| Baseline (specsheet)               | 0.3506        | —        |
| Consistency aggregation (NLI only) | 0.2792        | —        |
| UniLSTM (BERT-base)                | 0.4610        | 0.4481   |
| BiLSTM (BERT-base)                 | 0.4805        | 0.5130   |
| BiLSTM (DeBERTa-v3-base)           | 0.5130        | 0.4935   |
| BiLSTM (RoBERTa-base)              | 0.5390        | 0.5390   |

Table 3: Claim classification accuracy on the dev set.

| Class           | Precision | Recall | F1   |
|-----------------|-----------|--------|------|
| SUPPORTS        | 0.68      | 0.65   | 0.66 |
| REFUTES         | 0.57      | 0.44   | 0.50 |
| NOT_ENOUGH_INFO | 0.40      | 0.66   | 0.50 |
| DISPUTED        | 0.00      | 0.00   | 0.00 |
| Accuracy        | 0.54      |        |      |
| Macro avg       | 0.41      | 0.44   | 0.41 |
| Weighted avg    | 0.50      | 0.54   | 0.51 |

Table 4: Results on best model (RoBERTa-base + BiLSTM) on the dev set with retrieved evidence.

| Retrieval configuration         | Dev F-score |
|---------------------------------|-------------|
| Baseline (specsheet)            | 0.3378      |
| BM25 (full corpus)              | 0.1003      |
| BM25 + Re-ranker (full corpus)  | 0.1559      |
| BM25 + Re-ranker (curated pool) | 0.2465      |

Table 5: Evidence retrieval F-score on the dev set.

pending on the parameters chosen BiLSTM proved to perform better than the unidirectional network in almost all cases- BiLSTM achieved maximum 51.3% while UniLSTM - 46.1%. Because the two networks are identical apart from direction, this gain cannot be attributed to extra capacity; rather, reading the evidence sequence both forwards and backwards lets the aggregator condition each sentence’s contribution on the sentences that follow it as well as those that precede it. The gap between them widens further on gold evidence (51.30% versus 48.05%), indicating that the BiLSTM exploits clean evidence more effectively. Based on these results we decided to use BiLSTM for all subsequent classifications.

Holding the BiLSTM aggregator fixed, we then compare the three frozen encoders (Figure 2, Table 4). RoBERTa-base is the strongest, reaching 53.90% both on retrieved and on gold — a big improvement over BERT-base (46.10%) and a smaller improvement over DeBERTa-v3-base (51.3%). RoBERTa’s advantage over BERT is consistent with its longer pre-training over more data, which yields [CLS] representations that transfer

well to the climate domain without fine-tuning. In the Figure 2, we can see that DeBERTa-v3-base underperforms (overfits strongly as seen from the loss function over the epochs) despite being a more recent architecture; we attribute this to its disentangled-attention design, which produces no natural pooled representation. However, we noticed that even though RoBERTa performs better, DeBERTa seems to be more ‘fair’ towards the disputed claims, giving a better distributed and less overfitted result. However, the best overall configuration when considering straight up accuracy is RoBERTa-base with the BiLSTM aggregator which achieves 53.9% development accuracy, a 18.84-point absolute improvement over the specification-sheet baseline of 35.06%, and also clearly outperforms the standalone NLI consistency aggregator (27.92%).

### 3.5 Analysis

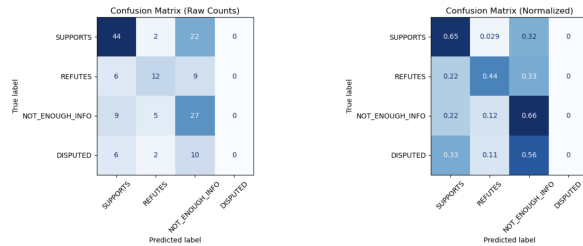


Figure 3: Confusion matrix for the best model (RoBERTa-base + BiLSTM) on the dev set of retrieved evidence.

BM25 alone over the full 1.2M corpus performed poorly (0.1003) because keyword matching struggles capturing with the “nuanced” language of climate claims where evidence is often indirect or paraphrased. By adding the Transformer re-ranker improved F-score by 55% (0.1003 > 0.1559) on the full corpus demonstrating that cross-encoder semantic scoring captures relevance beyond keyword that overlapped. We opt for switching to the curated evidence pool which further improved F-score to 0.2465 a 137% improvement over BM25 alone because the curated pool contains higher quality climate-specific passages directly related to training claims. The final retrieval F-score (0.2465) remains below the random baseline (0.3378) this is attributed to exact-match evaluation penalising semantically valid but non-annotated passages. In Figure 3 further reveals that our best model (RoBERTa-base + BiLSTM) correctly classifies most SUPPORTS and REFUTES claims but

fails entirely on the DISPUTED class.

## 4 Conclusion

In conclusion, we presented a hybrid retrieval and aggregation framework with our strongest configuration of RoBERTa-base model with BiLSTM aggregator that achieves 53.90% development accuracy with improvement of the baseline model with over 18.84 point and outperforms the NLI aggregator of 27%. It is found that contextualization between the evidence matters and affects the aggregator design choice of frozen encoder has a larger effect than the aggregator design, with RoBERTa transferring better to the climate domain. Future work should address the failure to predict the DISPUTED class even with our best model, and explore the joint training of retriever and aggregator on a larger climate corpus.

## Team Contributions

- **Karthik Kannan:** Literature review, system design and building the retrieval pipeline.
- **Ivan Smilenov:** Building encoder and classifier, running experiments and finalizing structure.
- **Ardelia Shaula Araminta:** Data analysis and report writing.

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